Holonomy, Finite Homogeneity, and the Nonidentification of Quantum Composition

A mathematical examination of prephysical-to-physical bridge proposals

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Abstract

We examine four bridge proposals using explicit hypothesis-to-conclusion bookkeeping. First, a geometric determinant identity holds on $\mathbb{C}\mathrm{P}^n$: for a nonorthogonal geodesic polygon, the complex determinant of Levi–Civita tangent holonomy is the $(n+1)$-st power of the normalized Bargmann invariant, relative to fixed connection and inner-product conventions. Second, real three-level pure-state geometry $\mathbb{R}\mathrm{P}^2$ has reflection monodromy despite local rotational holonomy, giving a concrete global discriminator from $\mathbb{C}\mathrm{P}^1$. Third, finite exactly homogeneous subsets of the Bloch sphere have covering radius at least $\arccos(59/60) \simeq 10.47531384°$; a published Gromov–Hausdorff theorem rules out the limit of finite homogeneous metric spaces being any $\mathbb{C}\mathrm{P}^n$. Fourth, even complete elementary-system complex quantum geometry and probabilities do not identify the composite state space: minimal separable and ordinary quantum composites are operationally different. These are conditional mathematical results and obstructions. None identifies a prephysical substrate, a clock, quantum dynamics, or the emergence of classical mechanics.

1 Conventions and limits

Let $H = \mathbb{C}^{n+1}$ with the inner product conjugate linear in the first slot and linear in the second. Rays $[\psi]$ of unit vectors form $\mathbb{C}\mathrm{P}^n$ with the standard Fubini–Study metric and natural complex structure J. We distinguish: *(i)* supplying this geometry as a mathematical premise, *(ii)* proving a relation inside it, and *(iii)* independently identifying that geometry through preparations and records. Only (i) and (ii) are performed below.

For rays $x_0, \ldots, x_{m-1}$ with nonzero consecutive overlaps, put $x_m = x_0$, choose normalized representatives, and define

$$B(x_0, \ldots, x_{m-1}) = \prod_{i=0}^{m-1} \langle \psi_i, \psi_{i+1} \rangle, \qquad b = B/|B| \in U(1). \tag{1}$$

Each edge is the unique shortest Fubini–Study geodesic. Orthogonal adjacent rays, with nonunique shortest geodesics, are excluded. $\mathrm{Hol}^{LC}(\gamma)$ transports $T^{1,0}_{x_0}\mathbb{C}\mathrm{P}^n$ around the indicated oriented geodesic polygon.

2 Exact determinant identity with its genuine inputs

Theorem 1 (Determinant and Bargmann product)**.** *Under the conventions above,*

$$\boxed{\det_{\mathbb{C}} \mathrm{Hol}^{LC}(\gamma) = b^{n+1}.} \tag{2}$$

For reversed traversal both sides are complex conjugated. The usual real determinant of tangent holonomy is instead identically $+1$*; the complex determinant uses* J*.*

Proof. Let $L \to \mathbb{C}\mathrm{P}^n$ be the tautological Hermitian line bundle with the connection obtained by orthogonal projection from the trivial flat H-bundle. Along a shortest nonorthogonal projective geodesic, a horizontal lift has endpoint phase chosen to make the overlap with the starting lift real and positive. Iterating the phase choices around the polygon yields line holonomy $h_L = b^{-1}$. Let $Q = H/L$ carry the quotient Hermitian connection. The standard Hermitian holomorphic bundle identities, respecting the induced Chern connections, give

$$T^{1,0}\mathbb{C}\mathrm{P}^n \cong \mathrm{Hom}(L, Q) = L^* \otimes Q, \quad \det Q \cong L^*, \quad \det T^{1,0}\mathbb{C}\mathrm{P}^n \cong (L^*)^{\otimes(n+1)}.$$

Because the Fubini–Study metric is Kähler, its Levi–Civita connection on $T^{1,0}\mathbb{C}\mathrm{P}^n$ equals the Chern connection. The determinant connection therefore has holonomy $h_L^{-(n+1)} = b^{n+1}$. The result presupposes a complex Hilbert space, a unitary inner product, its Fubini–Study geometry, and the induced connections. It does not derive these data from a relational prephysical model. □

Remark 2 (Metric and orientation caveat)**.** *Complex conjugation of every ray is a Fubini–Study isometry, preserves all transition magnitudes* $|\langle\psi, \phi\rangle|^2$*, but maps* $J \mapsto -J$*,* $\omega \mapsto -\omega$ *and* $b \mapsto \bar{b}$*. Thus an unoriented transition-probability metric alone does not canonically select a signed Bargmann phase. In particular, recovering* b *from the determinant as an* $(n+1)$*-st root has a root ambiguity; when* J *is not fixed, its complex determinant is not an intrinsically signed scalar. Complex-conjugate descriptions can be physically equivalent, so this observation is a representational nonselection, not evidence of distinct physics.*

Remark 3 (Relation to Schild's ladder)**.** *Schild's ladder is a* numerical approximation *to parallel transport, not an independent production mechanism for complex amplitudes. Guigui and Pennec [3] prove convergence for appropriately rescaled iterates; dividing only the base path without scaling transported vectors need not converge. The finite numerical observations ascribed to other directors are not substituted for this theorem or for a source inspection.*

3 A real three-level monodromy witness

Proposition 4 (Full holonomy of $\mathbb{R}\mathrm{P}^2$)**.** *Take* $\mathbb{R}\mathrm{P}^2 = S^2/(x \sim -x)$ *with its round quotient metric. Its restricted tangent holonomy is* $SO(2)$*, whereas the full tangent holonomy contains reflections and equals* $O(2)$*. The real commutant of the latter is* $\mathbb{R}I$*, while that of the former is* $\{aI + bJ : a, b \in \mathbb{R}\} \simeq \mathbb{C}$*.*

Proof. At $p = (0, 0, 1)$, follow the great semicircle $\gamma(t) = (\sin t, 0, \cos t)$, $0 \le t \le \pi$, from p to $-p$. The geodesic tangent $T(t) = (\cos t, 0, -\sin t)$ and $e_y = (0, 1, 0)$ are parallel. At $t = \pi$, these are $(-1, 0, 0)$ and $(0, 1, 0)$. The antipodal identification has differential $-I$, so the induced return map on the initial ordered basis (e_x, e_y) is $\mathrm{diag}(1, -1)$. This has determinant -1 and is a noncontractible

loop in $\mathbb{R}\mathrm{P}^2$. The local curvature generates all $SO(2)$ rotations from contractible loops. Hence the full holonomy is $O(2)$. Commuting first with all rotations yields $aI + bJ$; commuting with the displayed reflection forces $b = 0$. □

The standard pure-state space for a real three-dimensional Hilbert system is $\mathbb{R}\mathrm{P}^2$, while a complex two-dimensional Hilbert system has $\mathbb{C}\mathrm{P}^1 \cong S^2$. After matching constant curvature scales they are locally indistinguishable but globally differ by the above loop; this statement assumes those pure-state geometries, and is not a physical detection proposal.

4 Quantitative finite-carrier obstruction

For a finite $S \subset S^2$, let $R(S) = \sup_{x \in S^2} \min_{s \in S} d_{S^2}(x, s)$, with unit-sphere geodesic distance. Call S exactly homogeneous when the distance-isometry group of S acts transitively on S.

Theorem 5 (Uniform uncovered spherical cap)**.** *Every nonempty finite exactly homogeneous $S \subset S^2$ satisfies*

$$\boxed{R(S) \geq \arccos(59/60) = 10.475313843\ldots^\circ .}$$

Here 10.48° *is a* rounded approximation*, not a rigorous lower bound of exactly* 10.48°*. The inequality is not claimed sharp.*

Proof. If S lies in a proper linear subspace of $\mathbb{R}^3$, some point at distance at least $\pi/2$ from that subspace suffices. Otherwise every distance-preserving permutation of S extends uniquely to an orthogonal transformation of $\mathbb{R}^3$; the full symmetry group $\Gamma \leq O(3)$ is finite and transitive on S. Its orientation-preserving normal subgroup $\Gamma^+ = \Gamma \cap SO(3)$ has index at most two.

By the finite subgroup classification of $SO(3)$ [2], Γ^+ is cyclic, dihedral, tetrahedral, octahedral, or icosahedral. In the last three cases $|\Gamma^+| \leq 60$, hence $|S| \leq |\Gamma| \leq 120$. If $N = |S|$, spherical caps of radius r have area $2\pi(1 - \cos r)$. Covering the sphere requires $N 2\pi(1 - \cos R(S)) \geq 4\pi$, hence $R(S) \geq \arccos(1 - 2/N) \geq \arccos(59/60)$.

In the cyclic C_k cases with $k \geq 2$ and dihedral D_k cases with $k \geq 3$, the principal rotation axis is preserved as an unoriented axis by the full normalizing group Γ. The orbit S therefore lies in at most two latitude circles with colatitudes θ and $\pi - \theta$. Regardless of θ, a point on an appropriate latitude has colatitude distance at least $\pi/4$ from those circles, so $R(S) \geq \pi/4$. Low-order exceptional axial cases have $|\Gamma^+| \leq 4$, thus $|S| \leq 8$ and the area bound applies. This covers all cases. □

Corollary 6 (No exact-homogeneity continuum through GH limits)**.** *No sequence of finite homogeneous metric spaces can converge in Gromov–Hausdorff distance to $\mathbb{C}\mathrm{P}^n$ for any $n \geq 1$, with its Fubini–Study metric.*

Proof. Gelander [1] proves that the only manifolds obtainable as Gromov–Hausdorff limits of finite homogeneous metric spaces are tori. The compact simply connected $\mathbb{C}\mathrm{P}^n$ is not a torus. □

Remark 7. *The cap theorem concerns finite* embedded *exact-homogeneity orbits. Gelander's theorem is stronger but depends specifically on Gromov–Hausdorff convergence and exact finite metric homogeneity. Neither prohibits nonhomogeneous approximants, asymptotically homogeneous limits, or different nonmetric limit schemes.*

5 Composition remains unselected even with complete local quantum data

Theorem 8 (Elementary-system geometry does not identify quantum composites)**.** *There exist two consistent finite-dimensional operational state-space constructions with identical full elementary qubit state spaces* $\mathcal{S}_1 = \{\rho \succeq 0 : \operatorname{Tr}\rho = 1\}$*, identical local effects* $0 \preceq E \preceq I$*, and identical local reversible unitaries, hence identical* $\mathbb{C}\mathrm{P}^1$ *geometry, transition probabilities, Bargmann products and holonomy, but different two-system state spaces and distinguishable correlations.*

Proof. For the ordinary complex quantum composite set $\mathcal{S}_{AB}^{Q} = \{\rho_{AB} \succeq 0 : \operatorname{Tr}\rho_{AB} = 1\}$. For the minimal composite set

$$\mathcal{S}_{AB}^{\mathrm{sep}} = \operatorname{conv}\{\rho_A \otimes \rho_B : \rho_A, \rho_B \in \mathcal{S}_1\}.$$

Both contain all product preparations, are convex and compact, admit every standard local effect and independent local operation, and give the same statistics on every elementary-system experiment. One may consistently restrict global state-changing instruments in the minimal theory to separability-preserving instruments, including local CPTP instruments and measure-and-reprepare-separable instruments. An entangled state is not thereby silently preparable by a global instrument.

For the Bell ray $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$, the partial transpose of its projector has eigenvalues $\{-1/2, 1/2, 1/2, 1/2\}$. Thus it is not separable. Set $A_0 = Z$, $A_1 = X$, $B_0 = (Z + X)/\sqrt{2}$ and $B_1 = (Z - X)/\sqrt{2}$. With local dichotomic effects, the CHSH combination reaches $2\sqrt{2}$ on $|\Phi^+\rangle$ in the quantum composite. Every minimal-composite state is a convex mixture of products, so its CHSH correlations admit a local hidden-variable factorization and obey $|\mathrm{CHSH}| \leq 2$. The two operational constructions therefore differ, even though all their elementary-system geometry and statistics agree. □

This is a *nonidentification witness for an insufficient premise set*, not a disproof of quantum composition or of reconstruction theorems that additionally assume compositional axioms. The established generalized-probabilistic-theory distinction between minimal and other tensor products is discussed in [5].

6 Conditional classification and scope of third-order interference

Barnum, Müller and Ududec [4] require *three* single-system postulates (classical decomposability, strong symmetry, and absence of third-order interference) to obtain a restricted list of theories. No-third-order-interference *alone* does not establish that list. Even when a symplectic/holonomy test discriminates among the listed pure-state geometries, it does not supply the operational facts establishing the postulates, composition, dynamics, or an experimentally calibrated geometry. The real $\mathbb{R}\mathrm{P}^2$ example in Proposition 4 illustrates why full holonomy, not merely restricted holonomy, must be used if a global commutant criterion is proposed.

7 Research disposition and next mathematical obligations

1. Theorem 1 is a proved *conditional geometric identity*: it presupposes the very complex quantum geometric data a prephysical emergence must explain.

2. Proposition 4 is a concrete exact global obstruction to conflating local holonomy with global topology.
3. Theorem 5 and Corollary 6 constrain *exactly homogeneous finite metric* strategies; they say nothing universal about approximate/asymmetric finite models.
4. Theorem 8 is an explicit nonselection witness for composite physics given all elementary-system quantum data. Any total-ontology map must add and justify composition and admissible dynamics.
5. No prephysical primitive set, emergence of the Born rule, physical clocks, physical observations, or quantum-to-classical scaling limit is established. The parent Initial External Grounding stage stays open.

Reproducibility and assistance. A separate Python verifier checks 1,000 seeded Bloch-sphere triangle identities, the phase-conjugation symmetry, the one-loop reflection matrix, a Bell/PPT witness, and the decimal radius; these are finite checks, not proofs of the general theorems. This AI-assisted research examination was drafted by GPT-6 (OpenAI), acting as LLM Director B at the research direction of Kirtus Justus. The author remains responsible for scientific claims; the assistant's work is *internal research provenance*, not independent expert review or provider endorsement. The Director Z disposition was available as quoted text; its claimed G7 manuscript and Z1 archive were not independently supplied for inspection in this examination.

References

[1] T. Gelander, "Limits of finite homogeneous metric spaces," *L'Enseignement Mathématique* **59** (2013), 195–206. https://ems.press/content/serial-article-files/44250.

[2] H. T. A. Bui, "Classifying the finite subgroups of SO_3," University of Chicago REU notes (2020). https://math.uchicago.edu/~may/REU2020/REUPapers/Bui%2CAn.pdf.

[3] N. Guigui and X. Pennec, "Numerical Accuracy of Ladder Schemes for Parallel Transport on Manifolds," *Found. Comput. Math.* **22** (2022), 757–790, published online 2021. https://doi.org/10.1007/s10208-021-09515-x.

[4] H. Barnum, M. P. Müller, and C. Ududec, "Higher-order interference and single-system postulates characterizing quantum theory," *New J. Phys.* **16** (2014), 123029. https://doi.org/10.1088/1367-2630/16/12/123029.

[5] G. Eftaxias, M. Weilenmann, and R. Colbeck, "Multisystem measurements in generalized probabilistic theories and their role in information processing," *Phys. Rev. A* **108** (2023), 062212. https://doi.org/10.1103/PhysRevA.108.062212.

[6] E. M. Rabei, Arvind, N. Mukunda, and R. Simon, "Bargmann invariants and geometric phases: A generalized connection," *Phys. Rev. A* **60** (1999), 3397–3409. https://doi.org/10.1103/PhysRevA.60.3397.